



**NATIONAL INSTITUTE OF TECHNOLOGY
PUDUCHERRY**

**EE1113 – POWER ELECTRONICS LAB
MINI-PROJECT PROPOSAL**

**Power Factor Correction
Using Boost Converter**

Semester: *IV*

Department: *Electrical and Electronics Engineering*

Degree: *B. Tech*

Academic Year: *2026-2027*

By Batch A2:

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Objective:

The aim of this project is to design and implement an active power factor correction circuit to adjust the power factor by shaping the input current to a sinusoid and making it in-phase with the input supply.

Overview:

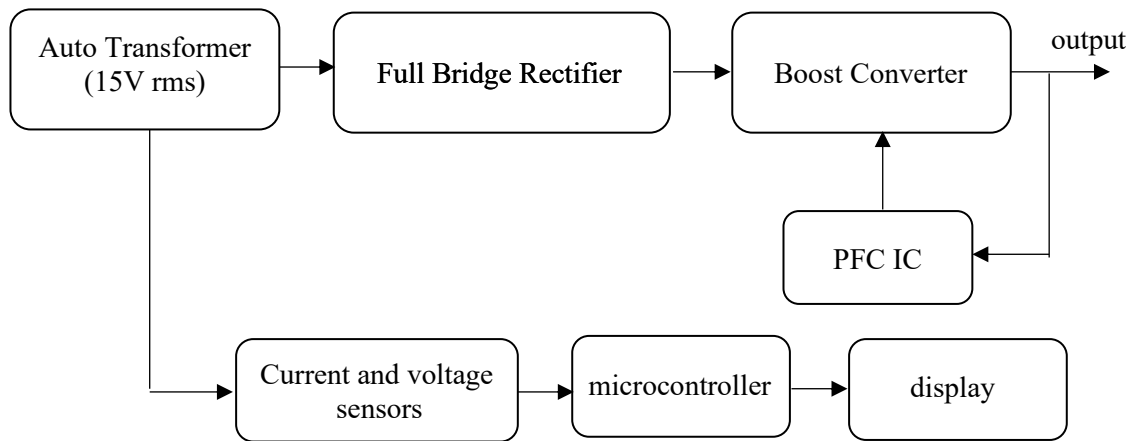
Power Factor Correction (PFC) is essential for the efficient utilization of electrical energy as a mismatch between voltage and current increases losses and causes overheating in distribution infrastructure. While passive PFC solutions are simple, they are often too bulky and inflexible for dynamic loads. Thus, an active PFC circuit is required.

For our project, we are taking a full-bridge rectifier for device that lowers the power factor which would act as an input to the power factor correction circuit.

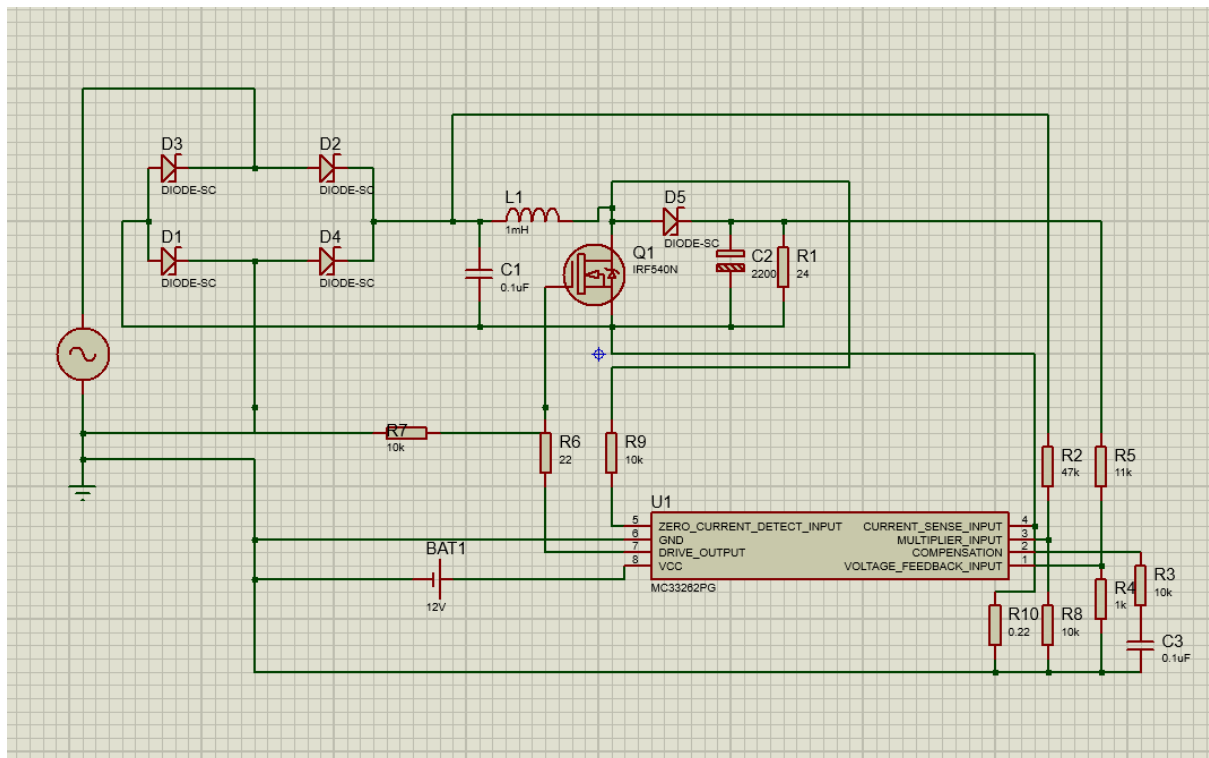
The main circuit consists of a boost converter whose gate signal is connected to a PFC IC which monitors the output voltage (essentially acting as a control loop). Here, the IC adjusts its duty ratio every cycle by comparing the actual inductor current to the reference sinusoid it generates from the rectified voltage for the circuit to provide a high power factor.

Finally, the power factor is then displayed on an LCD screen using an Arduino by calculating it from the voltage and current across the circuit.

Block Diagram:



Circuit Diagram:



Power Factor Correction Circuit

Components:

Sl. No	Component	Quantity	Specifications
1	Autotransformer	1	230V $\rightarrow$ 15VAC, $\geq$ 1A, 50Hz
2	Schottky Diodes	5	1N5822
3	Inductor	1	1–2 mH, $\geq$ 2A current rating
4	MOSFET(IRFZ44N)	1	55V, 49A, N-channel
5	Capacitor	1	2200 μ F, 50V, Electrolytic (polarized)
6	Ceramic Capacitor	2	0.1 μ F, 50V
7	Resistor	As Required	22 Ω , 10k Ω , 11k Ω , 1k Ω , 47k Ω & 10k Ω , 1/4W
8	PFC Controller IC	1	MC33262, DIP-8
9	Arduino Uno R3	1	-
10	LCD Display	1	16 $\times$ 2 LCD, 5V
11	Heat Sink	1	-
12	Connecting Wires	As required	-

Timeline:

Week 1	Attain all components
Week 2	Construct foundational circuit (with simulation)
Week 3	Implement the duty cycle aspect
Week 4	Add a display for power factor